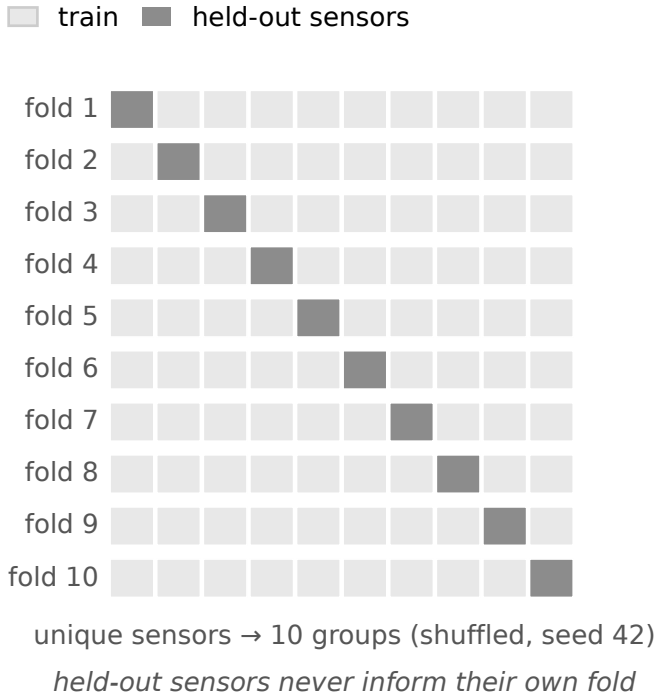
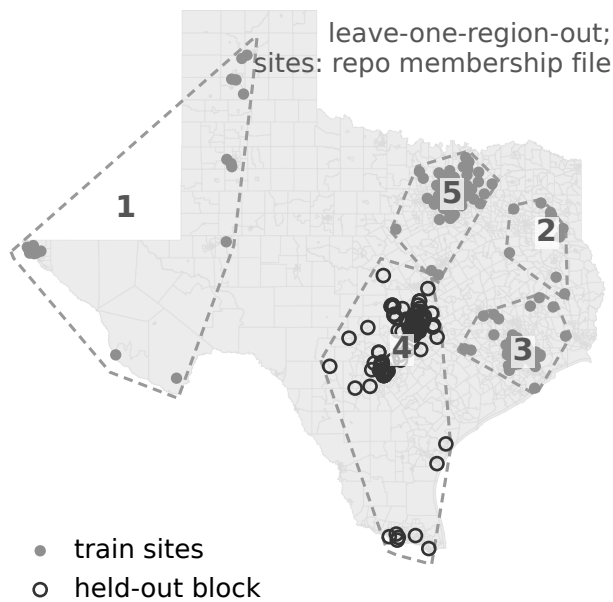


AQNet validation protocol — four axes plus the Tier-3 meta split

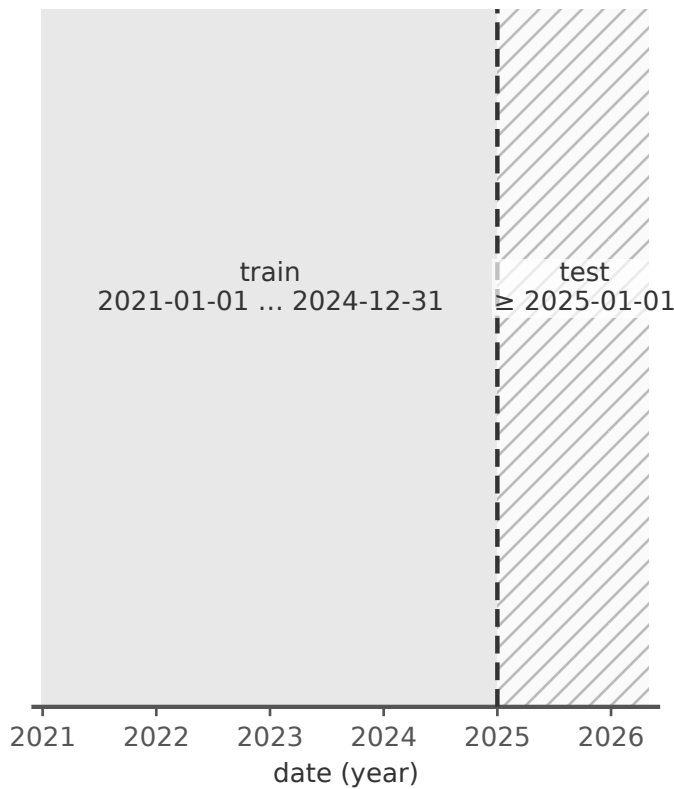
(a) LOSO — GroupKFold(10) over sensors



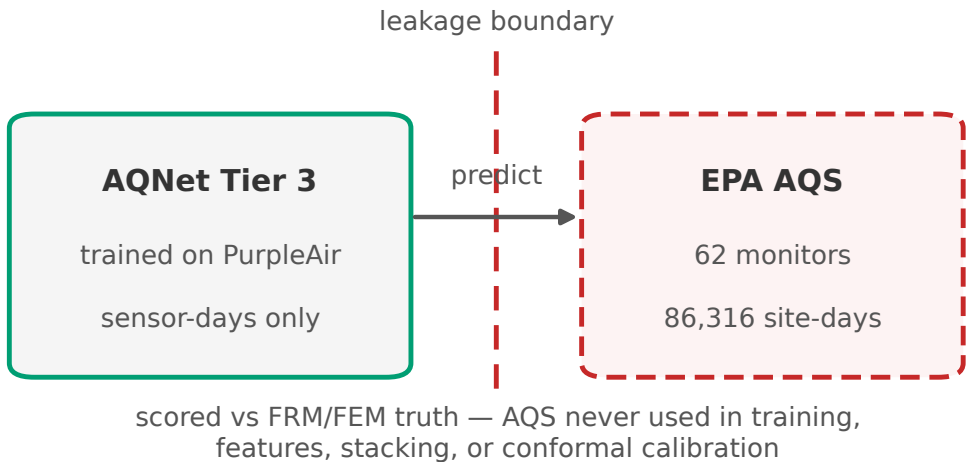
(b) spatial blocks — KMeans(5)



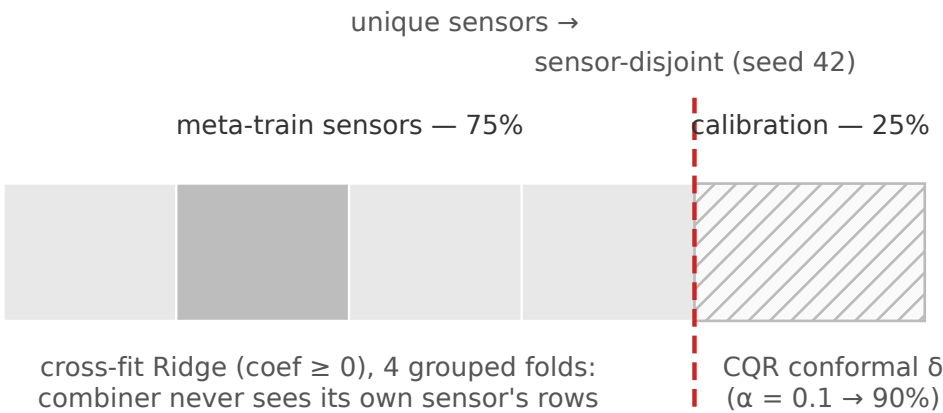
(c) temporal holdout — cutoff 2025-01-01



(d) external validation — EPA AQS FRM/FEM



(e) Tier-3 meta split — cross-fit vs calibration



one sub-fold at a time is held out inside meta-train (darker block)